Normal Distribution and Decision Errors

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MATH 240

Week 11, Lecture #1 — Ch. 13-14

Reminders

- HW7 due tonight at 11:59pm on Gradescope
- Exam #2 on Friday during lecture

Objectives

- **Calculate** z-scores to then find lower-tailed and upper-tailed probabilities using R
- **Find** percentiles for certain z-scores and data
- **Discuss** the impacts of type I and II errors based on discernibility level and sample size

Activity: Find the z-scores

For the following four situations, find the appropriate z-scores:

1. NBA teams spend about 14.2 seconds on average for each possession, with a standard deviation of 2 seconds. Say the Knicks spend 17 seconds for each possession.
2. It takes an average of 6 minutes to get to a class from home with a variance of 5 minutes. For one class, it takes 4.5 minutes.
3. For your bike rides, you average 10 miles per hour with a standard deviation of 1.5 mph. For this bike ride, you go 11.5 mph.
4. A sample of 20 students is collected to measure heights across campus, in which the pop. mean height is 67 inches and the standard deviation is 4 inches. In this sample, the average height is 64 inches and you are curious about the sample mean.

Finding Probabilities

As mentioned earlier, there is no closed-form solution to finding probabilities if we use integration on $f_X(x)$. As such, we introduce the following ways to find these probabilities:

- Empirical rule
- R (or other computation tools, not z-tables)

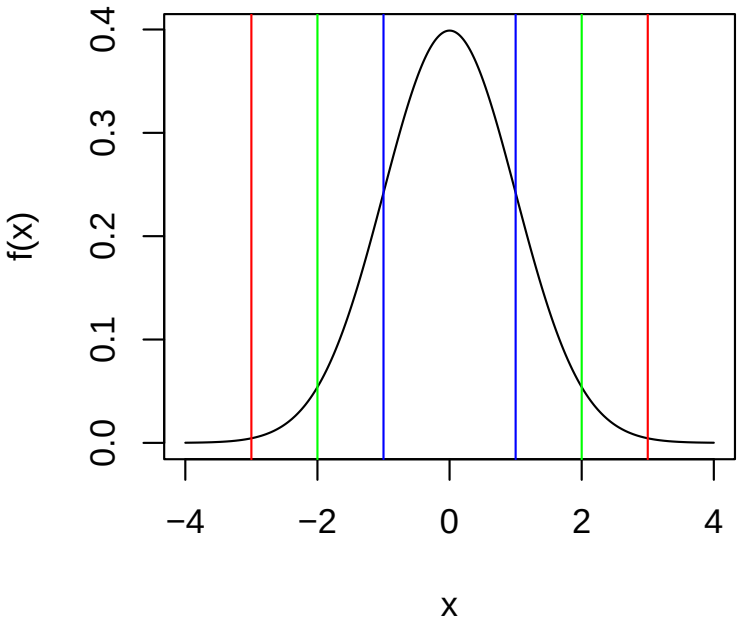
Different Probabilities

You may be tasked with finding the following probabilities:

- Lower-tailed: $\mathbf{P}(\mathbf{X} \leq \mathbf{x})$
- Upper-tailed: $\mathbf{P}(\mathbf{X} \geq \mathbf{x}) = 1 - P(X \leq x)$
- Between two values: $\mathbf{P}(\mathbf{a} < \mathbf{X} < \mathbf{b}) = P(X \leq b) - P(X \leq a)$
- More extreme than some value:
  $\mathbf{P}(|\mathbf{X}| \geq \mathbf{x}) = P(X \leq -|x|) + P(X \geq |x|)$

The Empirical Rule

- z-scores measure the standardized distance away from the mean
  - i.e. how many standard deviations away from mean
- 68-95-99.7
  - 68% of data falls within 1 SD of mean
  - 95% of data falls within 2 SDs of mean
  - 99.7% of data falls within 3 SDs of mean



Percentiles and z-Scores

Used for finding probabilities:

$$\begin{aligned} P(X \leq 15.6) &= P\left(\frac{X-\mu}{\sigma} \leq \frac{15.6-21}{5.4}\right) \\ &= P\left(Z \leq -1\right) \\ &\approx (1-0.68)/2 \\ &\approx 0.16 \end{aligned}$$

Used for finding quantiles:

$$0.84 = P(Z \leq z)$$

- $P(Z \leq -1) + P(-1 < Z < 1) = 0.16 + 0.68 = 0.84$
- $\implies z = 1$

Empirical Rule Probabilities

- At $z = \pm 1$:
  - $P(-1 < Z < 1) = 0.68$
  - $P(|Z| > 1) = 0.32$
  - $P(Z > 1) = 0.16$, $P(Z < -1) = 0.16$
- At $z = \pm 2$:
  - $P(-2 < Z < 2) = 0.95$
  - $P(|Z| > 2) = 0.05$
  - $P(Z > 2) = 0.025$, $P(Z < -2) = 0.025$
- At $z = \pm 3$:
  - $P(-3 < Z < 3) = 0.997$
  - $P(|Z| > 3) = 0.003$
  - $P(Z > 3) = 0.0015$, $P(Z < -3) = 0.0015$

Using R

- For other values of z (and the ones mentioned before): we can use the following R functions:
  - `pnorm(z)`: for lower-tailed probabilities
  - `qnorm(q)`: for percentiles

Using R

Suppose still that SAT scores $X \sim N(1020, 194^2)$. Calculate the percentile for 1300.

1. Find z:

$$z = \frac{1300 - 1020}{194} = 1.44$$

2. Use `pnorm()` to find lower-tailed probability:
   - `pnorm(1.44)` $= 0.9251$

Using R

Suppose still that SAT scores $X \sim N(1020, 194^2)$. Find the SAT score representing the 95th percentile.

1. Use `qnorm()` to find z of 92.51th percentile:
   - `qnorm(0.95)` $= 1.645$
2. Find x:

$$1.645 = \frac{x - 1020}{194} \implies x \approx 1339$$

Activity

Suppose a manufacturing plant packs containers of sugar and the average container has 500 grams of sugar with a standard deviation of 1.3 grams. Find the following:

1. Number of grams of sugar representing 1 and 2 SDs away from center
2. Probability of a container having less than 495 grams of sugar
3. Percentage of containers with over 505 grams of sugar
4. Probability of having containers between 498 and 501 grams of sugar